

Chapter 8

Tissue Microarrays from Frozen Tissues-OCT Technique

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Abstract

Tissue microarray technology is a new method used to analyze 100s to 1,000s of tumor samples on a single slide allowing high throughput analysis of genes and proteins on a large cohort. The original methodology involves coring tissues from paraffin-embedded tissue donor blocks and placing them into a single paraffin block. One difficulty with paraffin-embedded tissue relates to antigenic changes in proteins and mRNA degradation induced by the fixation and embedding process. We modified this technology by using frozen tissues embedded in OCT compound as donor samples and arraying the specimens into a recipient OCT block. Tumor tissue is not fixed before embedding, and sections from the array are evaluated without fixation or postfixed according to the appropriate methodology used to analyze a specific gene at the DNA, RNA, and/or protein levels. While paraffin tissue arrays can be problematic for immunohistochemistry and for RNA in situ hybridization analyses, this method allows optimal evaluation by each technique and uniform fixation across the array panel. Frozen tissue microarrays work well for DNA, RNA, and protein analyses, and may have significant advantages over the original technology for the assessment of some genes and proteins by improving both qualitative and quantitative results.

Key words: Frozen, Tissue microarray technology, Immunohistochemistry

1. Introduction

Recently developed high density tissue microarray technology involves arraying up to 1,000 cylindrical tissue cores from individual tumors on a tissue microarray (1). More than 200 serial sections can then be made from an individual microarray block and used for analysis of DNA, RNA, and/or proteins on a single glass slide. The technology is useful in that it allows rapid analysis of a large number of samples so that the statistical relevance of new markers can be determined in a single experiment. In addition, altered expression levels can be correlated to amplification

or deletion events in specific tumor samples using serial sections, allowing the simultaneous determination of gene copy number and expression analysis of candidate pathogenic genes and suppressor genes. Arrays have been made containing numerous tumor types (2) as well as multiple stages and grades within individual tumor types (3–5). This new technology has already proven useful for rapidly characterizing the prevalence and prognostic significance of differentially expressed genes identified using cDNA array technology (5–7) as well as genes involved in cancer development and progression (4, 5). Tissue microarrays have also been useful in identifying genes that are targets of chromosomal amplification (8, 9) as well as to study the expression patterns of putative tumor suppressor genes (10).

Some technical problems exist with the current methodology, however, relating to the fact that the arrayed samples have been prefixed and embedded in paraffin. The quality of the studies performed on sections from tissue array technology may be limited by the fixation methods used on the original sample. Buffered formalin solutions (and related compounds) are among the most widely used tissue fixatives. These chemicals fix the tissue by acting as progressive cross linkers between proteins and nucleic acids, by introducing modifications in RNA (adding monomethyl groups to its bases), and by producing coordinate bonds for calcium ions; these processes can damage RNA and alter target antigenic structure by blocking or damaging antibody binding sites (11, 12). Formalin fixation-induced alterations can make in situ analysis of DNA, RNA, and proteins suboptimal and variations in the duration of fixation can affect the quality and reproducibility of results (1, 12, 13). Fixation problems for FISH can be overcome by uniformly prefixing tissues in cold ethanol and embedding in paraffin (1) but this approach may not be optimal for array analysis of some proteins or for RNA using in situ hybridization. Paraffin embedding of ethanol-fixed tissue does not prevent RNA degradation (14). In addition, while ethanol fixation of tissue and subsequent paraffin embedding circumvents formalin fixation-related problems introduced by cross-linking, there are still problems relating to the embedding, and/or deparaffinization processes such as temperature-induced antigenic alterations introduced during the embedding process (12, 15, 16). One way to avoid these problems and ensure optimal preservation of antigens and nucleic acids is to use nonfixed (fresh frozen) tissue frozen at -70°C (17, 18). Herein we describe a change in the methodology that circumvents some of the problems with paraffin arrays and demonstrate that frozen tissue is suitable for creating tumor tissue microarrays.

2. Materials

1. Gentle Jane (Instrumedics Inc., Hackensack NJ).
2. Liquid Nitrogen.
3. Dry ice cylinders and sheets.
4. OCT Compound Embedding Medium (Miles, Inc. Diagnostic Division, Elkhart, IN).
5. Cryomold, Standard (Tissue-Tek disposable vinyl specimen molds 25 mm × 20 mm × 5 mm, Miles, Inc. Elkhart, IN).
6. Plastic Paraffin base mold (Simport Histosette II Biopsy Cassette from Fisher Scientific with lid removed).
7. Tissue microarrayer (Beecher Instruments, Silver Spring, MD).
8. Needles (1.0 mm, Beecher Instruments).
9. Basic CryoJane Tape Transfer System and slides (Instrumedics Inc., Hackensack NJ).

3. Methods

The following describes frozen tissue array technology from freezing of samples to using the array sections for IHC, FISH, and RNA in Situ applications. An example of a frozen tissue array is shown in Fig. 1.

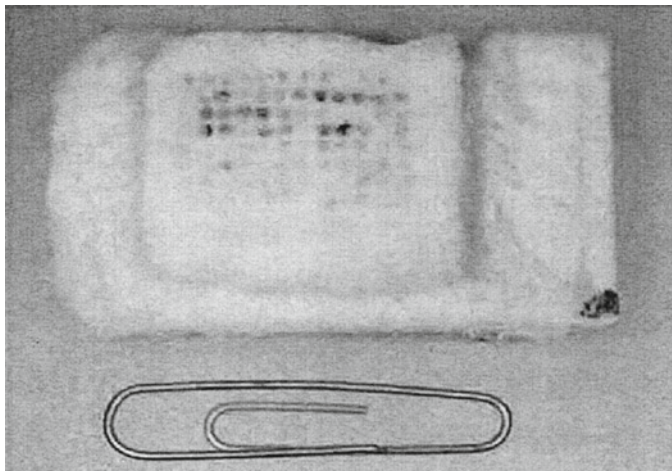


Fig. 1. Frozen tissue microarray. (Reprinted from *Am J Pathol* 2001, 159:1645–1650 with permission from the American Society for Investigative Pathology).

3.1. Preparation of Samples-Quick-Freezing and Embedding in OCT Compound

1. Procured tissue should be immediately frozen in liquid nitrogen or snap frozen using the Gentle Jane (Instrumedics Inc.) and stored at -70°C (see Note 1).
2. Prior to embedding samples, label molds with freezer pen and pour OCT compound into Cryomolds at room temperature.
3. Remove sample from -70°C freezer and place immediately on dry ice to avoid thawing as much as possible.
4. Pick up sample with prechilled tweezers and quickly push sample down to the bottom until completely submerged in OCT-compound filled Cryomold.
5. Immediately place on dry ice to freeze the embedded sample and avoid thawing as much as possible.
6. Store embedded samples at -70°C .

3.2. Preparation of OCT Recipient Block

1. Fill Cryomold completely and evenly with OCT compound at room temperature.
2. Snap off the base of a plastic paraffin mold, label it with a freezer pen, and make a mark at one corner (for proper orientation when making sections).
3. Place labeled base on top of OCT-filled cryomold and push gently so OCT compound seeps through and can be frozen to the plastic mold.
4. Place on flat surface in -70°C until frozen (approximately 10 min).
5. Once frozen remove plastic Cryomold.
6. Fix OCT Recipient Block in base of tissue arrayer and immediately surround with dry ice cylinders, replacing continuously when evaporated.
7. Make a mark with a freezer pen at 5 mm of your array biopsy needle so that when you punch it you know how deep to punch.
8. Starting a few millimeters from the edge of the block, remove a cylinder punch of OCT (approximately 1 mm in diameter by 5 mm deep) by pushing the needle down into the OCT as far as the mark and then releasing it to pull up the OCT cylinder. The removed OCT cylinder will immediately melt, so wipe the needle with a tissue to clean off the melted OCT, move at least 1 mm horizontally and make the next punch (see Note 2). Continue until you have finished the row. Move down at least 1 mm to make the next row.
9. Place a piece of dry ice on part of the block you are not punching to keep it frozen.
10. Make a recipient block with up to 120 holes in it (12×10) and store at -70°C until samples are ready to array.

3.3. Coring and Arraying Samples

1. Mount the OCT Recipient Block on the tissue arrayer and surround with dry ice cylinders to prevent melting.
2. Place OCT-embedded samples on dry ice. (Remove no more than 10 OCT-embedded samples from -70°C at a time because they will thaw slightly on dry ice).
3. Place OCT-embedded sample on the array bench and biopsy the sample by pushing the needle into the sample using slow pressure and turning the sample slightly from side to side until the needle has reached 5 mm (see Note 3).
4. Remove the biopsy from the sample and immediately hold a piece of dry ice to the needle to keep it from thawing.
5. Immediately return the sample to dry ice and remove the array bench (while keeping a cylinder of dry ice on the biopsy-filled needle).
6. Lower the needle to the appropriate hole on the array and push the biopsy into the hole while keeping a piece of dry ice next to it at all times to avoid thawing.
7. Move 1 mm to the next hole and repeat until the array is filled.
8. Make sure to keep the OCT Recipient block frozen at all times by surrounding continuously with fresh dry ice cylinders and dry ice cylinders on top of the completed rows to avoid thawing (see Notes 4 and 5).

3.4. Sectioning

Sections of the frozen tissue array must be made using the Instrumedics CryoJane Tape-Transfer System.

1. The array is mounted using OCT medium and fixed in the cryostat for sectioning.
2. Make note of the orientation of the array and the orientation of the sections on the slide.
3. Place slides and sectioning tape in the cryostat to cool.
4. Let array warm to the temperature of the cryostat -24 to -32°C .
5. Adjust and section the array until it is smooth and even.
6. Place sectioning tape over the array and use roller to make sure it is evenly sticking to array (see Notes 6 and 7).
7. Cut section.
8. Place section with tape fixed to it on cooled glass slide.
9. Use roller to gently roll section evenly onto slide.
10. Use crosslinker (part of CryoJane Transfer System) to link section to slide.
11. Gently peel off tape leaving section on slide.
12. Place slide with frozen section on dry ice and store slides at -70°C (see Notes 8–10).

4. Notes

1. The quality of the frozen tissue array is highly dependent on how the samples are originally frozen. We did a comparison on freshly removed rat tissues (rat liver tissue shown in Fig. 2) divided into three parts and subsequently frozen (a) in liquid nitrogen, (b) in the Gentle Jane (Instrumedics), and (c) on dry ice. The rat tissues were embedded in OCT, arrayed, sectioned, and HE stained to determine quality. We found tissue arrays made from tissue that was originally frozen more slowly on dry ice to be of poor quality, while tissue originally snap frozen in liquid nitrogen or using the Gentle Jane to be of similarly acceptable quality. Therefore, we find it best to snap freeze the tissue first, then embed the previously frozen sample in OCT compound on dry ice.
2. We prefer to prepare the OCT recipient block by coring holes in it before arraying the samples. This is because it is faster to make holes in the recipient block at a higher temperature when the OCT is a little softer and easier to core. The block can be made simultaneously with the array by coring a hole into the block, inserting the sample, and then coring the next hole, etc., but then care needs to be taken to keep the block completely frozen so the tissue stays frozen.
3. Unlike the paraffin array methodology, the same needle can be used for punching and coring. If a smaller needle is preferred (0.6 mm), be prepared to order extra needles as these needles bend and break easily when forced into frozen samples. The larger needles (1.0 mm or greater) are stronger but pressure should always be applied slowly and carefully to avoid bending or breaking the needle while coring the sample.
4. Even when the sample is surrounded by dry ice cylinders to keep it cold, if the tissue array is not made rapidly, the edges of the array may evaporate off a bit, leaving the array uneven

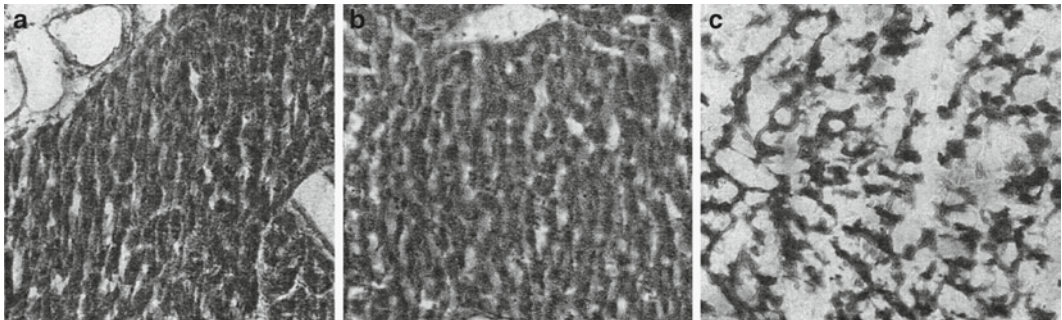


Fig. 2. Freezing methods test.

- and difficult to section. Therefore, it is best to center the samples and leave some space around the edges to avoid losing the samples on the edges.
5. The samples cannot be arrayed closer together than 1.0 mm because the OCT can crack when samples are not spaced properly.
 6. Unlike the paraffin arrays, you must use the same size needle for preparing the recipient block and coring the sample. Cores that are not fitting tightly into the recipient block can stick to the tape and get pulled out during sectioning.
 7. Sectioning can be difficult. The tape and block must be at the correct temperature or the tape will not stick to the block. Also, the tape may not stick to the OCT if it is not smooth, so the block must be cut until it is completely smooth and even, and then beautiful sections can be made. Thus it is important to push the cores all the way into the block when making the array so that you do not lose too much of your samples during the sectioning process.
 8. Most importantly, remember to keep the tissue, the array, and the slides frozen at all times to maintain quality until fixation.
 9. Remember samples in the frozen tissue array have not been fixed. Before performing IHC, FISH, or RNA in Situ, be sure to fix the array section in the appropriate fixation medium. Probes for FISH and antibodies for IHC are generally more efficient in frozen sections than paraffin, so the dilution of probe or antibody is usually less than recommended by suppliers who have usually optimized their technique for paraffin-embedded samples. For example, for antibodies that recommend 1:100 dilution, a 1:500 dilution may be sufficient.
 10. We have also made arrays of frozen cell lines and added cell lines as controls on the tissue array. There are two ways to do this. Cells can be pelleted and snap frozen in a microfuge tube in liquid nitrogen and then removed from the tube with a hammer and quickly embedded in OCT compound similarly to the tissue for subsequent coring. Alternatively, pelleted cells can be pipetted right into the OCT recipient block, but with this method they tend to freeze to the edges leaving a ring of cells with a hole at the center.

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